

Gating Benefits Depend on the Backbone: An Ablation Study of a Mamba-3 Block under Matched-Noise Cross-Condition Drift

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Abstract

The robustness of Mamba-style selective state-space models (SSMs) is often attributed to input-dependent selectivity. We test this attribution with a three-step ablation chain on one controlled testbed: a Mamba-3 block, bearing vibration signals (XJTU-SY, one speed-and-load transition), and noise-aware training at matched signal-to-noise ratio. Decision thresholds for each step were written down before the corresponding grid was run (internal pre-specification, not an external registry). First, replacing the input-dependent parts of Δ , A , B and C by learned constants—while the input-dependent trapezoidal weights and rotation angles of the Mamba-3 scan, the convolutional stem and the output gate are kept—costs at most about 5 pp under deep noise (paired intervals include zero), and the frozen block still outperforms a parameter-matched time-invariant S4D by 14–20 pp. Under the pre-specified criterion these four projections are therefore not the primary contributor. A second grid then freezes the trapezoidal weight and the rotation angles as well, leaving no tensor entering the scan dependent on the input: the advantage over S4D does not shrink but measures +21.9, +24.0 and +21.2 pp at 0, -2 and -6 dB (5/5 seeds, intervals excluding zero), with fewer parameters, so the block’s deep-noise advantage over a time-invariant SSM requires no input-dependent scan term at all. Second, removing the multiplicative SiLU gate from the frozen block costs 9–24 pp and returns it to roughly the level of S4D. Third, adding one gate branch to S4D recovers only 17–20% of the gap to the frozen block (the seed interval reaches the

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pre-specified 0.3 bound under last-epoch reporting); a host-by-gate contrast of 5–23 pp indicates that, in this design and training budget, the gate’s benefit depends strongly on the host. Removal and addition are not symmetric interventions, so this does not show that the gate cannot be transplanted. A separate, shorter per-class grid shows the gate-related differences concentrating on the minority inner-race class. Values are best-epoch macro-F1 on the evaluation condition; the central contrasts for four arms are re-assessed under last-epoch reporting. The evidence is conditional on one block, one condition pair with four bearings per side, five seeds and matched-noise training.

Keywords: state-space models, Mamba, input selectivity, gating, robustness, ablation, distribution shift, matched-noise training

1. Introduction

Selective state-space models (SSMs) of the Mamba family have rapidly become default sequence backbones, and a common narrative has formed around *why* they work: input-dependent selectivity—the ability to modulate state dynamics per token—is credited with filtering noise, adapting to distribution shift, and outperforming time-invariant predecessors such as S4D [1]. [2, 3] This belief motivates a growing family of “selective” architectures, yet it sits in unresolved tension with recent evidence: the first systematic S4D-versus-Mamba comparison on time-series classification finds the *simpler, time-invariant* model consistently superior [4], and theoretical analyses identify expressivity regimes where Mamba is provably weaker [5]. Which account is right—and, more importantly, *what exactly should be credited* when a Mamba-style block does prove robust? This paper addresses the second question for one block and one testbed, with a controlled ablation chain whose decision thresholds were fixed before each grid was run. The testbed is bearing vibration diagnosis under operating-condition drift and matched noise, in which the robustness gap is large (up to 18 pp), the evaluation labels are identical across all 865 cells (hash-verified), and the block can be dissected.

Our approach is deliberately *attributional* rather than *methodological*: we propose no new architecture and claim no state-of-the-art. Instead we take a regime in which a Mamba-style block exhibits a large, stable robustness advantage over a time-invariant SSM, and ask which ingredient earns it. The chain has three steps, each with adjudication thresholds fixed before the

25 corresponding grid was run:

- 26 1. **Test the classical selectivity terms (necessity test).** We con-
27 struct a *frozen* arm: the input-dependent parts of Δ , A , B and C
28 are replaced by learned constants. The Mamba-3 scan also takes an
29 input-dependent trapezoidal weight and input-dependent rotation an-
30 gles; these, the convolutional stem, the multiplicative SiLU gate, the s-
31 tate parameterisation, normalisation and projections are retained (Sec-
32 tion 3.2). If these four terms drive robustness, the frozen arm should
33 collapse toward the time-invariant baseline.
- 34 2. **Localise the necessary component.** On the frozen base, we remove
35 one component at a time (convolutional stem; SiLU gate; heavy-tailed
36 state parameterisation) under a single-variable discipline checked key-
37 by-key on the state dictionary (Section 3.2), and adjudicate each arm
38 against pre-specified Δ -thresholds.
- 39 3. **Test sufficiency by constructive grafting.** The component iden-
40 tified as necessary is added to plain S4D as a gate branch, and a pre-
41 specified recovery ratio decides whether “removal collapses” is matched
42 by “addition restores”.

43 The outcome is a conditional pattern. Freezing the input-dependent Δ ,
44 A , B , C projections costs at most about 5 pp in deep noise (paired intervals
45 include zero) while the block retains a 14–20 pp advantage over S4D (best-
46 epoch and last-epoch reporting, Section 4.5): under the pre-specified criterion
47 these projections are not the primary contributor. Freezing the trapezoidal
48 weight and the rotation angles as well—so that nothing entering the scan
49 depends on the input—does not shrink the advantage either: it measures
50 +21 to +24 pp over S4D with fewer parameters (Section 4.2). Removing the
51 SiLU gate costs 9–24 pp and takes the block to roughly the level of plain
52 S4D or below. Adding a gate branch to S4D recovers only $R = 0.17$ of
53 the gap (0.20 under last-epoch reporting; the seed interval reaches the pre-
54 specified 0.3 bound), with essentially no recovery at the deepest adjudicated
55 noise level. Removal and addition are not mirror-image interventions (they
56 differ in width, projection and parameter count), so we read the pair as
57 evidence that the gate’s benefit depends strongly on the host in this design,
58 not as proof that the gate cannot be transplanted. A per-class analysis on a
59 separate, shorter grid indicates that the gate-related differences concentrate
60 on the minority class.

Contributions. (i) A layered freezing experiment on a Mamba-3 block under distribution drift and matched noise, from the classical selectivity terms (Δ , A , B , C) through to a scan in which no tensor depends on the input, in which the block’s deep-noise advantage over S4D survives every step; each arm is verified input-independent by capturing the tensors the scan receives for two different inputs. (ii) Component-level evidence that the output gate is the one ablated component whose removal causes a large, one-signed loss, that one graft design recovers little of the gap, and a coarse host-by-gate interaction estimate. (iii) A reporting-rule sensitivity analysis (best epoch on the evaluation condition versus final epoch) showing that absolute levels shift by 1.4–9.5 pp while the central contrasts keep their sign. (iv) A documented pre-specification and review trail—evaluation-label hashes, thresholds written before each grid, and agent-assisted review—including the incidents in which automated adjudication departed from the pre-specified criteria; we release the trail.

2. Background and Related Work

2.1. Time-invariant and selective SSMs

Diagonal structured SSMs (S4D) [6] evolve a linear state with input-independent parameters (A , B , C , Δ), acting as a bank of learned linear filters. Mamba-style blocks [1] make Δ , B , C functions of the input token (selectivity), wrap the scan in a block containing a short causal convolutional stem, a multiplicative SiLU gate on a parallel branch, and residual/normalisation structure; the concrete block used throughout this paper follows the Mamba-3 formulation [7]. The narrative crediting selectivity for robustness draws on theory showing input-dependent dynamics can represent discontinuity-sensitive projections and counteract memory decay [8], and on [9]. We treat these as *hypotheses to be tested*, not established attribution: the theoretical results concern representability, which does not by itself determine which component earns empirical robustness in a trained block.

2.2. The unresolved comparison

Empirically the picture is contested. A systematic 59-dataset comparison finds S4D-based models superior to Mamba variants on clean time-series classification [4]; theoretical work identifies long-range regimes where Mamba is weaker than S4D [5]. Meanwhile, application literatures—including

95 machine-condition monitoring—contain many Mamba-hybrid methods re-
 96 porting noise robustness [10, 11, 12, 13, 14], but none, to our knowledge,
 97 isolates *which component* of the block is responsible under controlled con-
 98 ditions. Our clean-data results agree with the S4D-superiority finding (S4D
 99 matches or slightly exceeds the Mamba-style block on clean data). Whether
 100 the disagreement in the literature reflects a regime difference is not settled
 101 by a single testbed; what we can report is that, in the drift $\times$ noise regime
 102 studied here, the Mamba-style block’s advantage over S4D is not removed by
 103 freezing the classical selectivity terms.

104 2.3. Testbed rationale

105 We use bearing vibration diagnosis under cross-condition drift (speed
 106 and load both change) as the testbed, for three properties that matter to
 107 the attribution question rather than to the application: (a) the distribution
 108 shift is physically generated (a change of rotation speed and radial load be-
 109 tween recorded conditions), not simulated; (b) fault signatures are sparse
 110 transients—precisely the discontinuous structures selectivity theory predicts
 111 selective SSMS should excel at [8], which makes it a plausible arena for the
 112 selectivity hypothesis; and (c) the robustness gap between the Mamba-style
 113 block and S4D is large (up to 20 pp), giving the attribution chain high sig-
 114 nal. The testbed is two-class (details in Section 3.1); the scope of all claims
 115 is bounded accordingly (Section 6).

116 3. Experimental Protocol

117 3.1. Testbed, arms, and budget

118 Vibration windows from the XJTU-SY run-to-failure dataset [15] are split
 119 by operating condition: models train on one operating condition (37.5 Hz,
 120 11 kN) and evaluate on another (40 Hz, 10 kN), so speed and load both change
 121 (cross-condition), with additive white Gaussian noise applied at matched
 122 SNR to training and evaluation ($\{\text{clean}, +10, +6, 0, -2, -6, -10 \text{ dB}\}$). The
 123 task is binary fault classification (inner-race vs. outer-race dominant classes;
 124 class imbalance makes IR the hard class, Section 4.6). All arms share an
 125 identical training budget (50 epochs, 5 seeds; the per-class grid of Section 4.6
 126 used 20 epochs), identical splits, and identical evaluation labels, verified by a
 127 hash of the label vector: every reported cell carries `eval_sha = 6c20b367...`
 128 (Appendix A), recomputed independently during adjudication. The hash
 129 certifies the label vector, not the input windows. In total the study comprises

865 cells across five pre-specified grids (baseline grid 475; frozen-selectivity 105; component ablation 175; graft 35; per-class 75).

Composition of the data. Training uses the bearings of the 37.5 Hz/11 kN condition whose failure mode is inner race or outer race (Bearing2_1 is labelled IR; Bearing2_2, 2_4 and 2_5 are labelled OR); evaluation uses those of the 40 Hz/10 kN condition (Bearing3_3 and 3_4 are IR; Bearing3_1 and 3_5 are OR). Bearings with cage or mixed failure modes are excluded. Windows are non-overlapping segments of 2048 samples of the horizontal channel taken from the fault phase, which is delimited by a kurtosis/RMS onset criterion with a last-15% fallback; the evaluation set holds 3,504 OR and 1,744 IR windows, the training set 4,624 OR and 624 IR windows. Two consequences matter for interpretation. The IR class is learned from a single training bearing, so class-level results may reflect bearing-specific generalisation. And the five seeds resample initialisation, batch order and noise but not bearings: there are four independent bearings on each side, and none of the intervals reported here captures bearing-level uncertainty.

The principal arms are summarised in Table 1.

3.2. What the frozen arm freezes, and what the checks verify

In the Mamba-3 layer used here, one input projection emits the gate and value streams, the scan inputs B and C , the step-size and decay terms (Δ and A , through input-dependent additive terms), a trapezoidal weight and rotation angles. The frozen arm removes the input-dependent parts of Δ , A , B and C and replaces them with learned constants broadcast over time (for Δ the static bias of the original layer is kept; for A a new constant parameter plays the role of a static log-decay, passed through the layer’s own heavy-tailed clamp). The removed projections are deleted, so no parameter or gradient can leak. **The trapezoidal weight and the rotation angles remain functions of the current input**, as do the stem and the output gate. The frozen arm is therefore not a time-invariant scan: in the Mamba-3 formulation [7] the input-dependent rotation and trapezoidal weighting also enter the state update. Section 4.1 is accordingly a statement about the classical selectivity terms Δ , A , B , C only. Three further arms close that gap by the same device: *frozen-trap* replaces the trapezoidal weight by a learned per-head constant, *frozen-rot* replaces the rotation angles by learned constants (so the rotation becomes a learned position-only phase rather than a function of the current input), and *frozen-all* does both, leaving **no tensor**

Table 1: Principal experimental arms and parameter counts.

Arm	Description	Params
BM3	Mamba-style block stack (conv stem $k=7 \rightarrow$ selective scan $\rightarrow$ SiLU gate $\rightarrow$ projection), with and without an auxiliary physics loss (Section 3.4)	177,938
BM3-frozen	BM3 with the input-dependent parts of Δ , A , B , C replaced by learned constants; trapezoidal weight, rotation angles, stem, gate and the remaining components unchanged (Section 3.2)	112,410
S4D	Parameter-matched diagonal time-invariant SSM (LayerNorm; no BatchNorm anywhere in either SSM arm)	100,162
S4D-wide	S4D widened to $1.75\times$ parameters (capacity control)	178,498
frozen-conv	Single-component removal on the frozen base: conv stem (Section 4.3)	112,026
frozen-gate	Single-component removal on the frozen base: SiLU gate (Section 4.3)	79,642
frozen-A	Single-component removal on the frozen base: heavy-tailed A parameterisation (Section 4.3)	112,410
frozen-trap	Frozen base with the trapezoidal weight also made a learned constant (Section 3.2)	111,906
frozen-rot	Frozen base with the rotation angles also made learned constants	104,346
frozen-all	Frozen base with both also constant: no tensor entering the scan depends on the input	103,842
S4D+gate	Plain S4D plus the SiLU gate branch only (graft, Section 4.4)	116,802
CNN/TCN baselines	Conventional baselines for context	—

166 **entering the scan dependent on the input.** In each case the correspond-
 167 ing `in_proj` output slice is deleted. Section 4.2 reports them. For these arms
 168 the construction check adds the decisive probe: the tensors handed to the
 169 scan are captured for two different inputs through the same module, and a
 170 tensor the arm claims to have frozen is required to be bitwise identical across
 171 them while a tensor left selective is required to differ (all three arms pass).
 172 No kernel modification is needed, because the scan interface accepts these
 173 tensors regardless of their origin.

174 Single-variable discipline was checked by key-level state-dictionary diffs
 175 between two models built under the same seed. For the frozen arm the
 176 diff verifies that every non-target tensor has the same name and shape as in
 177 BM3; for the component-ablation and graft arms it additionally compares the
 178 values of non-target tensors at initialisation. Every differing key is mapped
 179 to an entry in an architecture map. We do not claim byte-level identity of
 180 non-target tensors in the frozen arm, and trained weights naturally differ
 181 between arms; the checks concern how the arms were constructed.

182 3.3. Reporting rule

183 The macro-F1 reported for each cell is the *maximum over epochs* of macro-
 184 F1 evaluated on the evaluation split, i.e., the held-out operating condition
 185 that every arm is scored on (the split covered by the fingerprint above).
 186 The harness defines no separate validation split, so the epoch is chosen with
 187 the evaluation data. Two consequences follow. First, absolute values are
 188 optimistically biased, by an amount that grows with an arm’s epoch-to-epoch
 189 variability. Second, a common rule cancels a shared offset between arms but
 190 not an arm-dependent one. The pre-specified thresholds were fixed against
 191 this rule, so all adjudications in Section 4 are reported under it, unchanged.
 192 Section 4.5 then re-assesses the central contrasts under a rule that does not
 193 use the evaluation split (the final epoch), and the boundaries section states
 194 what that re-assessment does and does not cover.

195 3.4. Confound elimination

196 Before component attribution, three candidate confounds were examined
 197 on the same grid. **Capacity:** S4D-wide ($1.75\times$ parameters) tracks S4D
 198 pointwise (mean -0.5 pp), so this particular increase in S4D’s parameter
 199 count does not explain the gap; S4D-wide was not re-run under last-epoch
 200 reporting, and the control does not address the change in channel mixing

201 that gate removal also entails. **Normalisation:** both SSM arms use Layer-
 202 Norm and contain no BatchNorm; the arm that degrades under noise is the
 203 one *without* any batch statistics, which argues against BatchNorm statistic-
 204 s mismatch as the explanation. Other normalisation inside the Mamba-3
 205 layer was not varied. **Auxiliary physics loss:** an exploratory kinematics-
 206 alignment regulariser carried by the original model contributes -0.1 pp point-
 207 wise (curves coincide); all attribution arms are therefore built on the pure
 208 cross-entropy variant.

209 3.5. Pre-specification and adjudication

210 Every adjudicated question in this study had numeric decision thresholds
 211 written to a task card and to disk before the corresponding grid was launched.
 212 This is internal pre-specification, not external registration: the supporting
 213 evidence is local file modification times and run-directory names (released,
 214 [Appendix A](#)), which are not tamper-evident and are not preserved by a
 215 repository checkout, so the trail lets readers inspect the order of events but
 216 cannot prove it. Later questions (component ablation, graft) were specified
 217 after earlier grids had been seen, as is natural in a staged design. Adjudi-
 218 cation was mechanical: an analysis pass applies the thresholds and emits a
 219 decision artifact; an independent reviewer pass re-computes aggregates, ver-
 220 ifies fingerprints row-by-row, and must cite file paths and modification times
 221 for every check. The pre-specified thresholds were binding in practice: two
 222 attempted post-hoc criterion substitutions and one stale-artifact false veri-
 223 fication were caught and re-adjudicated under the original rules — this is
 224 the one place where pre-specification actually bit. [Appendix C](#) details the
 225 incidents, the automated executor/reviewer implementation, and the human
 226 adjudication gates; all scientific decisions — thresholds, rulings, and interpre-
 227 tations — were made by the authors and are timestamped. Statistical tests
 228 (paired, per-condition) are reported throughout but were never adjudication
 229 criteria. A single repeat of one 2-epoch configuration in a separate process d-
 230 iffered by ~ 0.7 pp macro-F1 (non-deterministic kernel reductions; [Appendix](#)
 231 [B](#)); this is an observed repeat difference, not an estimated variability bound,
 232 and arm-level contrasts are judged against seed-level variability (2–9 pp).

233 4. Results

234 Throughout, “deep noise” denotes the pre-specified adjudication band
 235 $\{0, -2, -6 \text{ dB}\}$; -10 dB is reported but excluded from adjudication (the time-

invariant baseline is near chance there: a constant majority-class predictor scores 40.0% and a uniformly random predictor 48.6% macro-F1 on this evaluation set, against 52.3% for S4D). All values are macro-F1 (%), mean over 5 seeds, best-epoch on the evaluation condition (Section 3.3) unless stated otherwise. Figure 1 summarises the three-step ablation chain that structures this section.

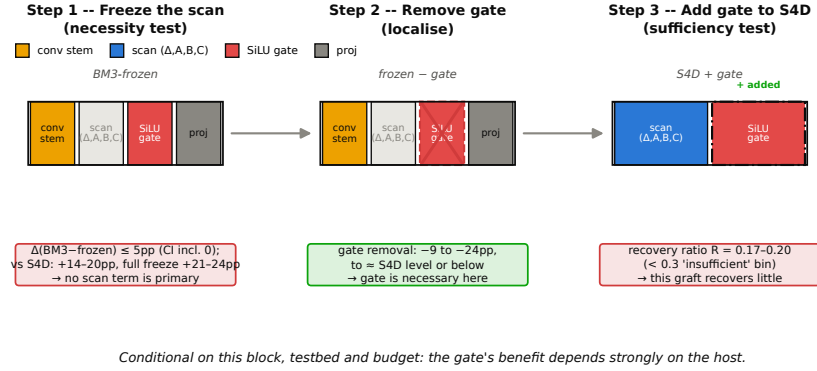


Figure 1: The three-step ablation chain with pre-specified thresholds: freeze the input-dependent Δ, A, B, C projections (Section 4.1), remove one component at a time on the frozen base to localise the necessary ingredient (Section 4.3), and graft that ingredient onto S4D to test sufficiency (Section 4.4).

4.1. Freezing the input-dependent Δ, A, B, C projections

Table 2: BM3 vs. BM3-frozen vs. S4D, macro-F1 (%) by SNR (frozen-selectivity grid). $\Delta(\text{BM3-frozen})$ and $\Delta(\text{frozen-S4D})$ in percentage points.

Condition	BM3	BM3-frozen	S4D	$\Delta(\text{BM3-frozen})$	$\Delta(\text{frozen-S4D})$
clean	78.9	78.9	81.0	–0.0	–2.1
0 dB	95.0	89.9	76.1	+5.1	+ 13.8
–2 dB	95.4	92.1	75.1	+3.3	+ 17.0
–6 dB	88.5	86.3	68.5	+2.2	+ 17.7
–10 dB*	68.1	64.3	52.3	+3.8	+12.0

Freezing the input-dependent Δ, A, B, C projections (Section 3.2) costs at most 5.1 pp anywhere in the adjudication band on the point estimates,

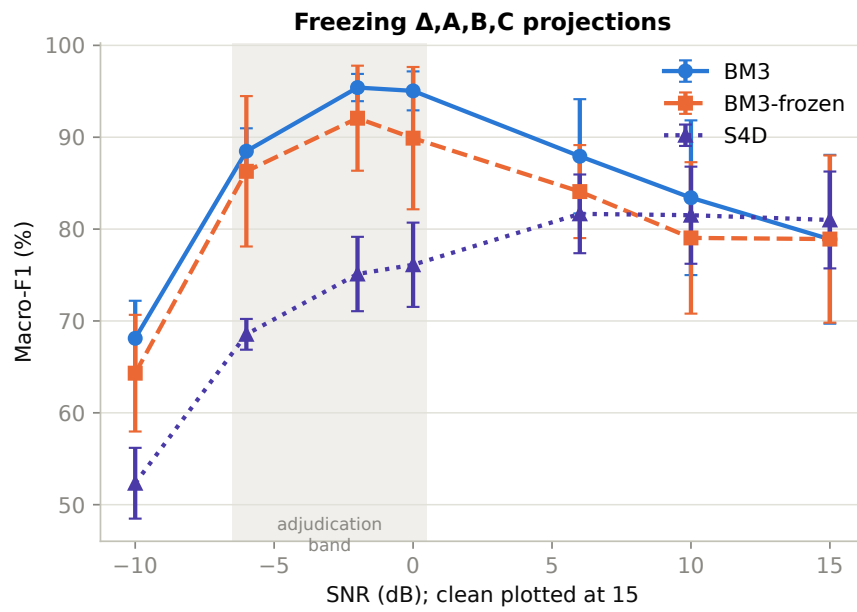


Figure 2: Macro-F1 vs. SNR for BM3, BM3-frozen, and S4D across the full noise grid. The frozen arm tracks BM3 closely and stays well above S4D throughout the adjudication band.

245 while the frozen block retains a 13.8–17.7 pp advantage over S4D (Table 2,
 246 Figure 2). Five seeds bound these quantities loosely: paired 95% intervals
 247 for $\Delta(\text{BM3-frozen})$ are $[-4.7, 15.0]$, $[-3.1, 9.8]$ and $[-7.5, 11.8]$ pp at 0, -2
 248 and -6 dB (BM3 ahead in 3/5, 4/5 and 2/5 seeds), and for $\Delta(\text{frozen-S4D})$
 249 $[-0.5, 28.0]$, $[5.5, 28.4]$ and $[5.8, 29.7]$ pp (5/5, 5/5, 5/5 seeds). The pre-
 250 specified rule attributed robustness to selectivity only if $\Delta(\text{BM3-frozen})$
 251 ≥ 10 pp in deep noise with the frozen arm collapsing toward S4D. Neither
 252 condition holds on the point estimates, so the pre-specified verdict is that
 253 these projections are *not the primary mechanism*. This is a failure to meet
 254 a confirmation criterion, not a demonstration that freezing is cheap: the
 255 intervals above are compatible with differences larger than 10 pp, and no e-
 256 quivalence margin was pre-specified or tested. One limit applies: the frozen
 257 arm still contains input-dependent trapezoidal weights and rotation angles,
 258 so this result concerns Δ, A, B, C only; Section 4.2 removes that limit. Un-
 259 der last-epoch reporting the re-run gives $\Delta(\text{BM3-frozen}) = +3.7, +0.9$ and
 260 $+0.2$ pp (best-epoch re-run: $+5.1, +3.5, +3.0$ pp; intervals still include zero
 261 at every level), so the point estimates fall further below the 10 pp threshold
 262 rather than rising towards it. On clean data S4D matches or slightly exceeds
 263 both Mamba-style arms—consistent with the systematic clean-TSC compar-
 264 ison [4]. In the noisy regime, the block’s advantage over S4D is not removed
 265 by freezing these four terms, so it must be carried by other parts of the block
 266 or by the remaining input-dependent scan terms.

267 4.2. Freezing the remaining input-dependent scan terms

268 Table 3 reports the three layered arms of Section 3.2 on the adjudication
 269 band (60 cells: 4 arms $\times$ 3 levels $\times$ 5 seeds $\times$ 50 epochs, the original training
 270 code, the same evaluation fingerprint, with the original BM3 arm re-run
 271 in the same grid). The question is whether the advantage that survived
 272 freezing Δ, A, B, C is carried by the two input-dependent terms the frozen
 273 arm retained.

274 It is not. With every tensor entering the scan input-independent, *frozen-all*
 275 is $+21.9, +24.0$ and $+21.2$ pp above S4D at 0, -2 and -6 dB under final-
 276 epoch reporting, positive in 5/5 seeds at every level and with intervals ex-
 277 cluding zero—if anything a larger margin than the partially frozen arm’s
 278 $+15.5, +19.8, +16.4$ pp (Table 7). None of the three freezing steps costs
 279 accuracy: the paired differences $\text{BM3-frozen} - \text{frozen-trap}$, $-\text{frozen-rot}$
 280 and $-\text{frozen-all}$ are negative at all three levels (between -1.8 and -6.3 pp
 281 under final-epoch reporting), i.e. the more heavily frozen arm is numerically

Table 3: Layered freezing of the scan’s input-dependent terms. Macro-F1 (%), mean over 5 seeds, under best-epoch and final-epoch reporting; Δ columns are paired differences under final-epoch reporting with 95% t -intervals over seeds.

Arm	0 dB		−2 dB		−6 dB	
	best	final	best	final	best	final
BM3 (re-run)	95.6	86.9	95.8	88.3	88.6	83.3
BM3-frozen (Δ, A, B, C)	90.6	83.1	92.3	87.3	85.7	83.1
frozen-trap	95.5	87.6	95.3	89.3	90.5	88.0
frozen-rot	94.6	86.7	94.6	89.1	89.0	86.4
frozen-all	96.5	89.5	96.8	91.5	90.5	87.9
S4D	75.9	67.6	75.1	67.5	68.6	66.7
<i>Paired differences, final-epoch reporting (pp)</i>						
frozen-all − S4D	+21.9 [16.7, 27.1]		+24.0 [21.4, 26.5]		+21.2 [17.5, 24.9]	
BM3-frozen − frozen-all	−6.3 [−17.6, 4.9]		−4.1 [−12.0, 3.7]		−4.8 [−17.8, 8.2]	
BM3 − frozen-all	−2.6 [−8.2, 2.9]		−3.2 [−7.0, 0.6]		−4.5 [−7.2, −1.9]	

282 ahead, with every interval including zero. The direction is not an artefact of
 283 capacity: *frozen-all* has 103,842 parameters against 112,410 for BM3-frozen
 284 and 177,938 for BM3. At the deepest level *frozen-all* also exceeds the full
 285 BM3 block by 4.5 pp ([−7.2, −1.9] for BM3 − frozen-all, 0/5 seeds in BM3’s
 286 favour); we report this but do not build on it, since five seeds and one speed
 287 pair cannot separate a genuine advantage of a time-invariant scan from a
 288 variance effect.

289 Two readings of the whole step are therefore available, and the data do not
 290 separate them: input-dependence of the scan contributes nothing measurable
 291 here, or it contributes something that the learned constants recover under this
 292 training budget. Either way the conclusion for the attribution chain is the
 293 same and now rests on a complete intervention rather than a partial one: **the**
 294 **Mamba-style block’s deep-noise advantage over a time-invariant**
 295 **S4D does not require any input-dependent term in its scan.** What
 296 remains to carry it is the rest of the block—the convolutional stem, the heavy-
 297 tailed A parameterisation, the normalisations and the output gate—which
 298 is what Sections 4.3–4.4 examine. The re-run also reproduces the original
 299 grids: the BM3 arm’s best-epoch means (95.6, 95.8, 88.6) exceed Table 2’s
 300 95.0, 95.4, 88.5 by 0.57, 0.43 and 0.16 pp (a different PyTorch/Triton build,
 301 Section 4.5).

Table 4: Single-component removal on the frozen base (component-ablation grid, trained separately from Table 2), $\Delta(\text{frozen} - \text{arm})$ in percentage points.

Component removed	0 dB	−2 dB	−6 dB	Adjudication
SiLU gate	+13.1	+21.1	+23.5	PRIMARY (all ≥ 8 pp)
conv stem ($k=7$)	−6.6	−2.7	+7.9	outside pre-specified bins (regime trade-off)
heavy-tailed $A \rightarrow$ S4D-style A	−2.4	−2.2	−4.2	outside bins (removal numerically beneficial; intervals include 0)

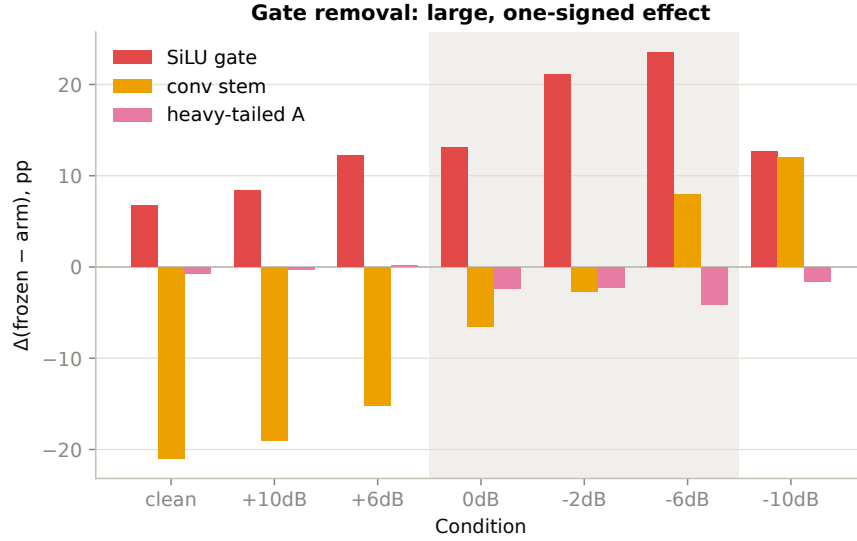


Figure 3: Component-removal Δ (percentage points, frozen – arm) by SNR for the three single-component ablations on the frozen base. Only gate removal is monotone, large, and one-signed across the adjudication band.

302 *4.3. Removing the gate is the only ablation with a large, one-signed effect*
303 “Necessary” is used here in the operational sense of this ablation: remov-
304 ing the gate from the frozen block, with the other components and the budget
305 unchanged, removes most of the block’s advantage. The gate result is large,
306 one-signed and monotone in noise depth: removal costs 13.1, 21.1 and 23.5 pp
307 at 0, −2 and −6 dB (paired 95% intervals [6.5, 19.6], [3.8, 38.5], [8.5, 38.6] pp;
308 5/5 seeds at every level). The gateless arm also gives up most of the Mamba
309 advantage: it sits **below plain S4D** at −2 dB (71.7 vs. 75.2) and −6 dB
310 (62.5 vs. 68.6), though not at 0 dB (77.6 vs. 76.2), and the gateless–S4D gap
311 itself is not resolved by five seeds (Section 4.5: under last-epoch reporting
312 the gateless arm is below S4D at −6 dB only). We read this as “at best on
313 par with S4D, and worse at the deepest noise level”, which suggests, but
314 does not establish, that components interact rather than add. The other two
315 components tell regime stories rather than robustness stories: the convolu-
316 tional stem helps by 7.9 pp at −6 dB (removal gives 78.1 vs. 86.0; paired 95%
317 interval of the difference [−2.1, 17.9] pp, 4/5 seeds) but *costs* 21.0 pp on clean
318 cross-speed data (removal yields 98.8 vs. 77.8, with seed-std collapsing from
319 8.1 to 0.6 pp)—consistent either with speed-specific local temporal pattern-
320 s mistransferring across conditions, or with optimisation friction under the
321 matched budget; we report both readings. Replacing the heavy-tailed state
322 parameterisation by an S4D-style one is numerically beneficial at all three
323 deep-noise levels (+2.4, +2.2, +4.2 pp), but every paired interval includes
324 zero (e.g., [−15.4, 7.1] pp at −6 dB), so we draw no directional conclusion
325 from it (numerically, the S4D-style variant is the best deep-noise cell in the
326 component grid, 90.1 at −6 dB).

327 *4.4. Adding a gate branch to S4D: limited recovery*

Table 5: Grafting the SiLU gate branch onto plain S4D (S4D unchanged key-by-key; only the gate branch added). The S4D+gate column is the graft grid; the S4D and BM3-frozen columns, and hence R , are the component-ablation grid of Section 4.3, as pre-specified, which is why they differ from Table 2 by 0.1–0.7 pp (independent runs of the same arms).

Condition	S4D+gate	S4D	BM3-frozen	Recovery ratio
0 dB	79.8	76.2	90.6	0.249
−2 dB	79.4	75.2	92.8	0.239
−6 dB	69.2	68.6	86.0	0.035

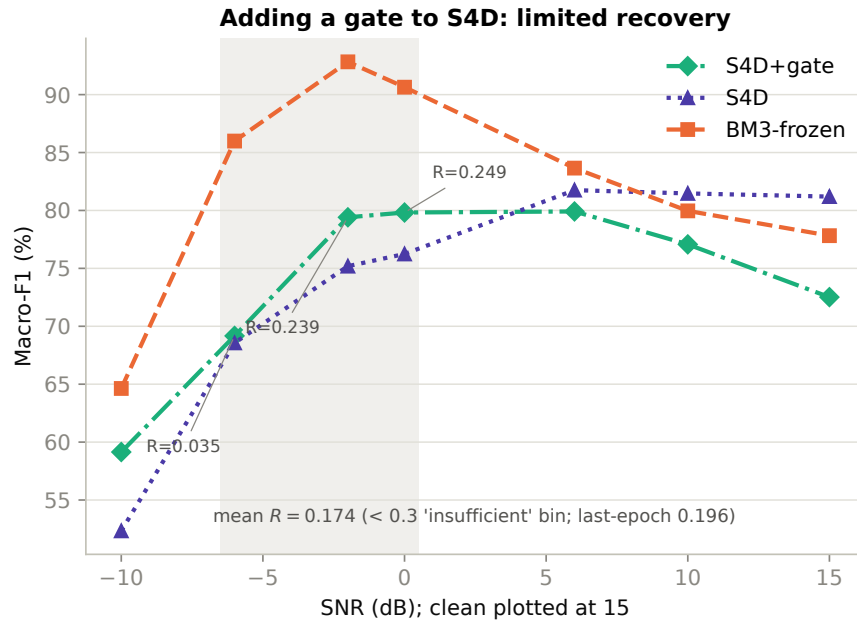


Figure 4: Graft outcome: S4D+gate, plain S4D, and BM3-frozen across the adjudication band, with the per-condition recovery ratio $R = (\text{S4D+gate} - \text{S4D}) / (\text{frozen} - \text{S4D})$ annotated. Recovery is near zero at the deepest adjudicated noise level.

328 Pre-specified recovery ratio $R = \text{mean over the band of (S4D+gate -}$
 329 $\text{S4D})/(\text{frozen - S4D}) = \mathbf{0.174}$. The pre-specified bins were $R \geq 0.7$ (con-
 330 firmed), $0.3 \leq R < 0.7$ (partial recovery) and $R < 0.3$ (insufficient); $R =$
 331 0.174 falls in the last. Resampling seeds within each arm gives a 95% in-
 332 terval of $[0.04, 0.29]$ for R , and the per-level paired gains of S4D+gate over
 333 S4D are $+3.6$ $([-2.2, 9.3])$, $+4.2$ $([1.1, 7.3])$ and $+0.6$ pp $([-2.2, 3.4])$. At the
 334 deepest adjudicated level the graft recovers essentially nothing. Removal
 335 and addition are not mirror-image interventions. In BM3 the gate has its
 336 own $64 \rightarrow 128$ projection and multiplies the scan output at the inner width
 337 of 128 before the output projection; removing it deletes 32,768 parameters.
 338 In the S4D host the grafted gate is a $64 \rightarrow 64$ linear branch (with bias)
 339 computed from the layer input and multiplying the SSM output before the
 340 residual addition; it adds 16,640 parameters over four layers. The two share
 341 the multiplicative SiLU form but not width, projection structure or param-
 342 eter count. Removal costs 13–24 pp in the frozen block whereas this graft
 343 adds 0.6–4.2 pp to S4D. What this shows is that *this gate branch, in this host*
 344 *and under this budget, recovers little*. Three readings fit the data and cannot
 345 be separated here: (i) host dependence, i.e. the gate needs the stem or scan
 346 context it co-trained with; (ii) implementation differences in width, projec-
 347 tion and placement; (iii) transplant friction, i.e. the grafted branch is harder
 348 to optimise in a foreign host under the matched budget (training converges
 349 normally, but convergence is not robustness). Separating them would need
 350 a width-matched graft, an equal-capacity non-multiplicative branch as con-
 351 trol, and a tuning or longer-budget comparison selected on a source-domain
 352 validation set; none was run.

353 The four arms do allow a coarse host-by-gate contrast, $I = [F(\text{frozen}) -$
 354 $F(\text{frozen-gate})] - [F(\text{S4D+gate}) - F(\text{S4D})]$, where F is macro-F1. Under
 355 last-epoch reporting $I = 5.2$ $[2.3, 8.1]$, 8.9 $[-6.2, 23.9]$ and 22.6 $[2.9, 42.4]$ pp
 356 at 0, -2 and -6 dB (seed-matched 95% t -intervals; positive in 5/5, 4/5 and
 357 5/5 seeds); under best-epoch reporting in the re-run it is 7.8 $[1.4, 14.2]$, 14.4
 358 $[-1.4, 30.3]$ and 23.1 $[8.7, 37.4]$ pp. A positive I means the gate adds more
 359 on the frozen host than on S4D. Because the host differs in several respects
 360 at once and the two gates are not aligned, I cannot be assigned to a specific
 361 component or interaction term, and the seed pairing across arms is nominal
 362 (arms share seed indices, not initialisations).

363 4.5. Sensitivity to the reporting rule

364 The tables above use best-epoch macro-F1 on the evaluation condition
 365 (Section 3.3). To measure how much this rule shapes the conclusions, we
 366 re-ran the four arms that carry the chain (BM3-frozen, S4D, frozen-gate,
 367 S4D+gate) over the adjudication band: 60 cells, 5 seeds, 50 epochs, the
 368 original training code and the same evaluation fingerprint, recording macro-
 369 F1 at every epoch and reporting it at the final epoch, where the cosine
 370 learning-rate schedule has annealed to zero. This rule uses no evaluation data
 371 to pick a model. The re-run used a different PyTorch/Triton build from the
 372 original grids; its best-epoch values reproduce the original arm-by-condition
 373 means to within -0.6 to $+1.4$ pp (the largest deviation is in the high-variance
 374 frozen-gate arm at -2 dB). The S4D-wide capacity control and the conv-
 375 stem and A -parameterisation ablations were not re-run; the original BM3
 376 arm was re-run in the stage-2 grid of Section 4.2, so $\Delta(\text{BM3-frozen})$ is
 377 available under both rules.

Table 6: Macro-F1 (%), mean over 5 seeds, under best-epoch reporting (re-run) and final-epoch reporting.

Arm	0 dB		-2 dB		-6 dB	
	best	final	best	final	best	final
BM3-frozen	90.6	83.1	92.3	87.3	85.7	83.1
S4D	75.9	67.6	75.1	67.5	68.6	66.7
frozen-gate	77.7	74.0	73.1	71.7	61.9	60.5
S4D+gate	80.9	71.5	79.9	74.3	69.2	66.6

378 Final-epoch values are lower than best-epoch values by 1.4 – 9.5 pp across
 379 the twelve arm-by-level means, so the absolute levels in Tables 2–5 are opti-
 380 mistic. The reduction is of similar size for the arms being contrasted (e.g.,
 381 BM3-frozen -7.4 , -5.0 , -2.6 pp and S4D -8.3 , -7.6 , -1.9 pp at 0 , -2 ,
 382 -6 dB), and the contrasts that carry the argument are correspondingly sta-
 383 ble (Table 7).

384 *Selectivity.* The frozen block’s advantage over S4D is $+15.5$, $+19.8$ and
 385 $+16.4$ pp (and $+21.9$, $+24.0$, $+21.2$ pp for the fully frozen scan, Section 4.2),
 386 versus $+14.6$, $+17.2$ and $+17.1$ pp under best-epoch reporting in the re-run.
 387 It remains positive in $5/5$, $5/5$ and $4/5$ seeds, with intervals excluding zero
 388 at every level. *Gate.* Removing the gate costs 9.1 , 15.6 and 22.6 pp (12.8 to

Table 7: Paired differences under final-epoch reporting, in percentage points: mean, 95% t -interval over seeds, and number of seeds (of 5) with a positive difference.

Contrast	0 dB	−2 dB	−6 dB
	+15.5	+19.8	+16.4
BM3-frozen − S4D	[4.3, 26.8] (5/5)	[12.0, 27.7] (5/5)	[4.1, 28.7] (4/5)
	+9.1	+15.6	+22.6
BM3-frozen − frozen−gate	[2.9, 15.3] (5/5)	[−0.8, 32.0] (4/5)	[5.8, 39.4] (5/5)
	+6.4	+4.2	−6.2
frozen−gate − S4D	[−2.9, 15.8] (4/5)	[−14.5, 22.9] (3/5)	[−23.5, 11.2] (1/5)
	+3.9	+6.8	−0.1
S4D+gate − S4D	[−0.8, 8.6] (4/5)	[2.4, 11.1] (5/5)	[−4.0, 3.9] (2/5)

23.8 pp under best-epoch reporting in the re-run), above the 8 pp pre-specified threshold at every level; the interval at −2 dB includes zero ([−0.8, 32.0] pp, 4/5 seeds) because of the large seed variance of the gateless arm (standard deviation 14.6 pp at −2 dB). *Gateless versus S4D*. Under final-epoch reporting the gateless arm is not below S4D at 0 and −2 dB (point estimates +6.4 and +4.2 pp) and is below it at −6 dB (−6.2 pp); all three intervals include zero. The statement that gate removal “falls below S4D” is therefore supported only as a point-estimate observation at the deepest level, and we phrase it as “to or below” elsewhere. *Graft*. The recovery ratio is 0.249, 0.341 and −0.003 per level and $R = 0.196$ on average (best-epoch re-run: 0.220), again in the pre-specified “insufficient” bin on the point estimate. The seed-resampled 95% interval is [0.09, 0.31] and so reaches the 0.3 boundary: with five seeds the data support “far below confirmation (0.7) and probably below 0.3”, not a precise value of R . The graft gain over S4D is resolved at −2 dB only (+6.8 pp, [2.4, 11.1]) and is indistinguishable from zero at −6 dB.

Because the final epoch of the same evaluation condition has been inspected repeatedly during this study, last-epoch values are also not a fresh confirmatory test. The reporting rule therefore changes absolute levels and the strength of the gateless-versus-S4D statement, but not the ordering BM3-frozen > {S4D+gate, frozen−gate, S4D}, the size of the frozen block’s advantage over S4D, or the verdicts of the pre-specified gate and graft rules.

4.6. The interaction effect concentrates on the hard class

Per-class analysis (a separate 75-cell grid at 20 epochs with best-epoch selection on the evaluation condition: 5 arms $\times$ {clean, 0, −6 dB} $\times$ 5 seeds,

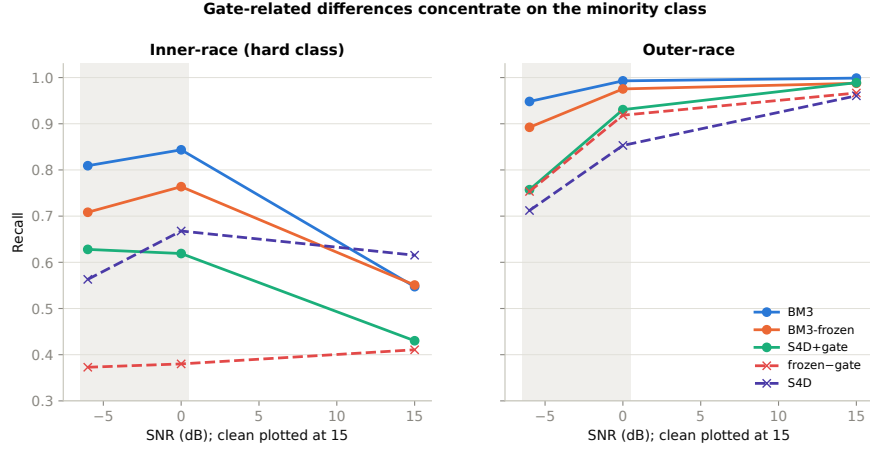


Figure 5: Per-class recall (inner-race, IR, and outer-race, OR) as a function of noise depth, split by gate presence. Gated arms lift IR recall under matched noise; gateless arms let it deteriorate.

per-sample predictions dumped and re-verified against reported metrics cell-by-cell): the inner-race class is hard from the outset (clean recall 0.41–0.62 vs. 0.96–1.00 for outer-race). Its trajectory under noise is governed by gate presence: in all three gated arms IR recall at -6 dB exceeds its clean value (e.g., $0.55 \rightarrow 0.81$ for BM3), while in both gateless arms it is below its clean value (frozen–gate: $0.41 \rightarrow 0.37$; S4D: $0.62 \rightarrow 0.56$, non-monotone, rising to 0.67 at 0 dB) (Figure 5). Both ablation directions hit IR hardest—gate removal costs IR 2.3–3.8 $\times$ more F1 than OR at every level (F1 from predictions pooled over seeds), and the graft’s noise-time gains also concentrate on IR (-6 dB: +6.8 pp IR vs. +4.2 pp OR). Because macro-F1 averages the two classes, the arm-level robustness gaps of Sections 4.1–4.4 are carried substantially by the hard class. The gate-related differences are therefore class-concentrated rather than uniform, and fall on the class with the lowest recall. We did not measure classification margins, so we do not describe IR as a “low-margin” class. Because this grid is separate from the main 50-epoch grid and the IR class comes from a single training bearing (Section 3.1), these results are descriptive and do not decompose the main-grid effects.

5. Discussion

What can be credited? The ablation chain supports a narrow, conditional statement: within this Mamba-3 block, on this testbed and with

noise-aware training, (i) making the scan fully input-independent—first the Δ , A , B , C projections, then the trapezoidal weight and rotation angles as well—did not remove the block’s deep-noise advantage over a time-invariant S4D, and at the deepest level the fully frozen arm matched or exceeded the original block with fewer parameters, (ii) removing the output gate removed most of that advantage, and (iii) adding one gate branch to S4D recovered little of it. These observations are consistent with a host-dependent gate benefit. They do not identify which other component the gate depends on. Nor do they separate “input-dependence contributes nothing here” from “what it contributes is recovered by learned constants under this budget”; the two readings are indistinguishable in our data and lead to the same attribution. Representability results for selectivity [8] and the S4D-superiority finding on clean data [4] are not contradicted by a single ablation on one testbed; our clean-data parity agrees with the latter, and our deep-noise result suggests that in this block the output gate, rather than any input-dependent term of the scan, accounts for most of the measured advantage.

Implications for backbone selection and for transplant practice. For practitioners choosing backbones for noisy, drifting signal regimes, the results suggest that the block’s benefit should be judged as a composition, and that a gate added to a different host should not be expected to reproduce it without a width-matched and tuned implementation (our graft was one untuned design, Section 4.4). A hypothesis for future work follows: a recomposed block guided by the component map (no conv stem, gate retained, S4D-style A) may match or exceed the original on this testbed; we leave it untested, because post-hoc recomposition after seeing the map would not carry the evidential status of the pre-specified chain.

Why might gating be composition-bound? We refrain from a mechanistic claim beyond the data, but note three compatible observations. First, the gate-related differences concentrate on the minority, lowest-recall class (Section 4.6), and matched-noise training lifts that class *only when the gate is present*—suggesting the gate is the component through which matched-noise training is converted into hard-class margin, i.e., an exploiter of noise-aware training rather than a test-time denoiser (this reading is bounded by the matched-SNR scope, Section 6). Second, its role appears closer to adaptive per-sample feature suppression than to generic denoising. Third, the gateless composition falling to or below S4D at the deepest noise levels is compatible with the other components being calibrated to a gated regime (e.g., the stem may pass features that are only safe when the gate can veto them); this is the

471 weakest of the three observations, since the gateless-S4D gap is not resolved
 472 with five seeds (Section 4.5). All are testable in future work via pairwise
 473 interaction grids and mismatched-noise protocols.

474 6. Boundaries and Limitations

475 We state the scope plainly.

- 476 1. **Testbed scope.** All claims are bounded to bearing vibration from one
 477 condition pair of one dataset (37.5 Hz/11 kN $\rightarrow$ 40 Hz/10 kN; speed and
 478 load both change), binary classification, GPU execution (no edge/real-
 479 time claims). Training has three OR bearings and one IR bearing;
 480 evaluation has two of each, and the five seeds do not resample bearings
 481 (Section 3.1). Additional condition pairs, independent bearings, multi-
 482 class settings and a second dataset are future work.
- 483 2. **Attribution level.** Credit assigned to “the rest of the composition”
 484 is by elimination within the enumerated component set; pairwise in-
 485 teraction terms are not decomposed, and the host-by-gate contrast of
 486 Section 4.4 is a four-arm estimate that cannot be assigned to a specific
 487 component.
- 488 3. **The transplant conclusion is budget-conditional.** All arms share
 489 a matched 50-epoch budget; the grafted gate trains normally yet re-
 490 covers little, and convergence is not robustness (Section 4.4)—but we
 491 cannot exclude that a longer or graft-tuned budget would recover more.
 492 Our statement is therefore limited to *this gate branch, in this host, un-
 493 der a matched training budget*.
- 494 4. **Matched-SNR scope.** Training and evaluation share the same in-
 495 jected SNR throughout (Section 3.1); “robustness” in this paper is
 496 matched-noise robustness under noise-aware training, not clean-train/noisy-
 497 test generalisation. The gate’s benefit may operate by exploiting matched-
 498 noise training (consistent with Section 4.6) rather than by test-time de-
 499 noising; attribution under mismatched-noise protocols is untested and
 500 may differ.
- 501 5. **Graft dual reading.** Interaction dependence vs. transplant friction
 502 (Section 4.4), the latter now subsumed under the budget condition of
 503 item 3.
- 504 6. **Noise-regime coverage.** Under strong coloured noise at -10 dB a
 505 TCN baseline slightly overtakes the Mamba-style block, and -10 dB

- 506 is near-chance for the time-invariant baseline and excluded from ad-
507 judication; the demonstrated advantage regime is matched, Gaussian-
508 dominated noise.
- 509 7. **Conv-stem dual reading.** Its clean-data cost admits both representation-
510 mistransfer and optimisation-difficulty readings, reported in place (Sec-
511 tion 4.3).
 - 512 8. **Reporting rule.** Reported values are best-epoch macro-F1 on the e-
513 valuation condition, with no separate validation split (Section 3.3); ab-
514 solute levels are optimistic by 1.4–9.5 pp in the arms re-run under final-
515 epoch reporting (Section 4.5). That re-assessment covers four arms at
516 the three adjudication levels only; the BM3 arm, the capacity control,
517 the stem and A ablations, the +10/+6/clean/−10 dB levels and the
518 per-class grid were not re-run, and no source-domain validation rule
519 was available. The final epoch is a fixed-budget rule, not a tuned one.
 - 520 9. **Scope of the freezing intervention.** The chain’s first step freezes
521 the input-dependent parts of Δ , A , B , C only; the fully input-independent
522 scan is a separate grid (Section 4.2) on the three adjudication levels,
523 which was specified after the earlier grids had been seen. No equivalence
524 margin was pre-specified for the cost of freezing, so “no measurable
525 cost” means the point estimates favour the frozen arms with intervals
526 including zero, not a demonstrated equivalence. Freezing is implement-
527 ed inside this Mamba-3 layer; a conclusion about input-dependence in
528 general would need other blocks and tasks.
 - 529 10. **Asymmetric gate interventions.** Gate removal (BM3) and gate
530 addition (S4D) differ in width, projection structure and parameter
531 count (Section 4.4); a width-matched graft and an equal-capacity non-
532 multiplicative control were not run.
 - 533 11. **Strength of the pre-specification evidence.** Thresholds were pre-
534 specified internally; the supporting evidence is local file times and run-
535 directory names, not an external time-stamped registry (Section 3.5).
536 The process-level repeat check is a single observation.
 - 537 12. **Construction checks.** For the frozen arm the state-dictionary check
538 compares names and shapes of non-target tensors; value-level compar-
539 ison was applied to the component-ablation and graft arms only.
 - 540 13. **Sample size.** Every cell mean uses five seeds. The smallest attainable
541 one-sided Wilcoxon p is 0.031, paired intervals are wide, and several
542 secondary statements (conv stem, A parameterisation, gateless versus

543 S4D) are not resolved at this size; conclusions rest on effect sizes that
544 clear the pre-specified thresholds, not on significance tests, which were
545 never adjudication criteria.

546 A single-repeat check of process-level non-determinism (~ 0.7 pp) and the
547 label-hash/adjudication trail are documented in the reproducibility appen-
548 dices ([Appendix A](#)–[Appendix B](#)).

549 7. Conclusion

550 In this Mamba-3 block, on one controlled drift $\times$ noise testbed with
551 matched-noise training, freezing the input-dependent Δ , A , B , C projections
552 costs at most about 5 pp (an estimate that five seeds leave imprecise) while
553 14–20 pp of advantage over a time-invariant SSM remains, and freezing the
554 trapezoidal weight and rotation angles as well—leaving no input-dependent
555 tensor in the scan—costs nothing measurable and leaves 21–24 pp of ad-
556 vantage with fewer parameters. The multiplicative output gate is the one
557 ablated component whose removal causes a large, one-signed loss (9–24 pp,
558 returning the block to roughly the level of plain S4D or below), yet one
559 gate branch added to S4D recovers only 17–20% of the gap (best-epoch and
560 last-epoch reporting). Because removal and addition are not matched inter-
561 ventions, we read this as a strong host dependence of the gate’s benefit in
562 this design, concentrated on the minority class, rather than as a proof of non-
563 transplantability. The evidence rests on one condition pair with four bear-
564 ings per side, five seeds and an evaluation-condition epoch-selection rule that
565 we re-assessed for four arms. We offer the three-step chain—freeze, remove,
566 add—as a protocol for turning architecture comparisons into component-level
567 tests, with the caveat that the interventions must be matched in width and
568 capacity for the add step to be conclusive, and we release our adjudication
569 trail, including the incidents in which automated adjudication departed from
570 the pre-specified criteria.

571 CRediT authorship contribution statement

572 Yi Wang: Conceptualization, Methodology, Software, Validation, Formal
573 analysis, Investigation, Data curation, Writing – original draft, Writing –
574 review & editing, Visualization, Supervision, Project administration.

575 **Declaration of competing interest**

576 The authors declare that they have no known competing financial inter-
577 ests or personal relationships that could have appeared to influence the work
578 reported in this paper.

579 **Declaration of Generative AI and AI-assisted technologies in the**
580 **writing process**

581 During the preparation of this work the authors used LLM-based agents
582 to execute and cross-check pre-specified analysis and adjudication passes as
583 described in Appendix C, and to assist with manuscript drafting and figure
584 generation. After using these tools, the authors reviewed and edited the con-
585 tent as needed and take full responsibility for the content of the published
586 article. All numeric thresholds, adjudication rulings, and scientific interpre-
587 tations reported in this paper were fixed and made by the human authors,
588 as detailed in Appendix C.

589 **Data availability**

590 The code, pre-specification artifacts, and adjudication trail for this study
591 are available at <https://github.com/jeff-1980/bm3-jizi>. The reposi-
592 tory includes the two follow-up grids of Sections 4.2 and 4.5 (120 cells with
593 per-epoch curves, their drivers, the layered-freeze arms and their construc-
594 tion check) and the harness’s own imported modules, so the training entry
595 point runs from a fresh checkout.

596 **Appendix A. Fairness protocol**

597 *Appendix A.1. Evaluation-set fingerprint*

598 Every reported cell in this study is tagged with `eval_sha256`, a SHA-
599 256 hash of the raw held-out test-label byte array, computed once per run
600 and asserted identical across every arm/condition/seed cell within a grid.
601 It fingerprints *that the same held-out label vector was scored every time*, not
602 model weights or predictions. It is also the split on which the best-epoch rule
603 of Section 3.3 selects epochs. The full value shared by all 865 cells across all
604 five grids is:

605 6c20b367522ce5db59720120f8a7cd12792c7e070c6ae84bd1985511998727de

606 This value is recomputed independently at adjudication time (not merely
607 copied from the run log) and cross-checked row-by-row against every grid’s
608 fairness report before any comparison across arms is treated as valid.

609 *Appendix A.2. Per-arm parameter counts*

610 Table A.8 lists every arm’s parameter count, each obtained directly via
611 `sum(p.numel() for p in model.parameters())` on the instantiated model
612 (not estimated), cross-verified against an independently maintained archi-
613 tecture map.

Table A.8: Per-arm parameter counts (verified against `arch_map.md` and harness source).

Arm	Parameters
BM3 (<code>bm3_kin</code> / <code>bm3_nokin</code>)	177,938
BM3-frozen (<code>bm3_frozen</code>)	112,410
S4D (<code>s4d</code>)	100,162
S4D-wide (<code>s4d_wide</code> , capacity control, $1.75\times$)	178,498
frozen-conv (<code>frozen_noconv</code>)	112,026
frozen-gate (<code>frozen_nogate</code>)	79,642
frozen-A (<code>frozen_As4d</code>)	112,410
S4D+gate (<code>s4d_plus_gate</code>)	116,802

614 *Appendix A.3. Five-batch pre-specification timeline*

615 Table A.9 lists, for each of the five pre-specified grids, the artifact that
616 fixed its adjudication thresholds, that artifact’s creation/modification date,
617 and the corresponding grid’s run-directory start time. In every case the
618 prereg artifact predates the grid it governs.

619 Two honest caveats. First, the baseline batch predates the orchestration
620 infrastructure used for the other four batches by five days; its prereg trail
621 rests on the task card’s creation timestamp alone, not on the same automat-
622 ed executor/reviewer machinery documented in [Appendix C](#), and we state
623 this rather than implying a uniform process across all five batches. Second,
624 for the component-ablation grid, two additional JSON files bearing “pre-
625 reg” in their name exist on disk (`secondary_prereg_20260704-0708` and
626 `secondary_deltapp_prereg_20260704-0719`) that *post-date* the grid’s start
627 time; both are explicitly marked void in the corresponding decision artifact’s

Table A.9: Pre-specification predates data collection for all five grids.

Grid (cells)	Canonical prereg artifact	Prereg date	Grid start
Baseline (475)	Task-tracking-system card (<code>created:</code> 2026-06-27, mtime 15:02)	2026-06-27	2026-06-27 17:31
Frozen-selectivity (105)	<code>frozensel_prereg.json</code> (m- time 18:26:54)	2026-07-02	2026-07-02 18:27
Component ab- lation (175)	Task-tracking-system card (<code>created:</code> 2026-07-03), materialised in <code>prereg_provenance.md</code>	2026-07-03	2026-07-03 10:34
Graft (35)	<code>graft_prereg_provenance.md</code> (mtime 21:20:36)	2026-07-05	2026-07-05 22:41
Per-class (75)	Task-tracking-system card (<code>created:</code> 2026-07-06)	2026-07-06	2026-07-07 21:21

628 `void_preregs` field, and neither was used to adjudicate any result in this pa-
629 per. They are the on-disk residue of the first post-hoc criterion substitution
630 described in [Appendix C](#), retained here (rather than deleted) as part of the
631 released adjudication trail.

632 Appendix B. Confusion matrices and a repeat-run check

633 Appendix B.1. Full confusion-matrix set

634 Table [B.10](#) gives the complete confusion-matrix set underlying the per-
635 class analysis of [Section 4.6](#): 5 arms $\times$ 3 conditions (clean, 0 dB, -6 dB),
636 each summed over 5 seeds (support OR= 17,520, IR= 8,720 per cell). These
637 counts were dumped per-sample and re-verified against the reported recal-
638 l/precision/F1 metrics cell-by-cell before use.

Table B.10: Full confusion-matrix set (summed over 5 seeds; support OR=17,520, IR=8,720 per cell). Columns give true $\rightarrow$ predicted counts.

Arm	Condition	OR $\rightarrow$ OR	OR $\rightarrow$ IR	IR $\rightarrow$ OR	IR $\rightarrow$ IR
BM3	clean	17,500	20	3,948	4,772
	0 dB	17,395	125	1,364	7,356

Arm	Condition	OR→OR	OR→IR	IR→OR	IR→IR
BM3-frozen	−6 dB	16,613	907	1,664	7,056
	clean	17,303	217	3,917	4,803
	0 dB	17,089	431	2,060	6,660
S4D+gate	−6 dB	15,634	1,886	2,543	6,177
	clean	17,327	193	4,969	3,751
	0 dB	16,303	1,217	3,322	5,398
frozen−gate	−6 dB	13,267	4,253	3,244	5,476
	clean	16,935	585	5,139	3,581
	0 dB	16,092	1,428	5,406	3,314
S4D	−6 dB	13,200	4,320	5,469	3,251
	clean	16,828	692	3,354	5,366
	0 dB	14,945	2,575	2,897	5,823
	−6 dB	12,477	5,043	3,810	4,910

639 *Appendix B.2. Repeat-run check*

640 Because two independent process runs of an identical configuration are
641 not bitwise reproducible on this hardware (non-deterministic GPU kernel re-
642 ductions in the selective-scan and normalisation ops), we measured a process-
643 level noise floor directly: the same arm, condition, and seed (`bm3.frozen`,
644 clean, seed 0, 2 epochs) was run twice in separate processes with identical
645 evaluation fingerprints and identical parameter counts confirmed. The two
646 runs’ best macro-F1 differed by

$$|0.69025 - 0.69712| = 0.00687 \approx 0.687 \text{ pp},$$

647 which we report as an observed repeat difference of ~ 0.7 pp (cited in Sec-
648 tion 3.5). It is a single repeat of one arm, one condition and a 2-epoch
649 run, and therefore neither an estimate nor a bound of process variability for
650 full-length runs. It is an order of magnitude below the seed-level variabil-
651 ity observed within the deep-noise band across the curve tables underlying
652 Figures 2 and 4 (standard deviations typically 2–9 pp, e.g. BM3-frozen 5.7–
653 9.4 pp and S4D 1.7–4.5 pp at $\{0, -2, -6\}$ dB). Arm-level effects in Section 4
654 are judged against seed-level variability, not against this single repeat.

655 Appendix C. Adjudication incidents, automation disclosure, and 656 human adjudication gates

657 This appendix gives the material Section 3.5 summarises in one sentence:
658 what the pre-specification discipline actually caught, how the analysis/review
659 pipeline was implemented, and exactly where human judgement entered. We
660 report this in full rather than in summary because the growing use of au-
661 tomation in pre-specified analysis pipelines raises exactly the failure modes
662 documented below, and because the reader should be able to audit, not mere-
663 ly take on faith, that no post-hoc criterion change survived into the results
664 reported in Section 4.

665 Appendix C.1. Automation disclosure

666 The mechanical stages of this study’s analysis pipeline — applying pre-
667 specified numeric thresholds to aggregated results and emitting a decision
668 artifact (the “executor” role), and independently re-computing aggregates,
669 re-verifying evaluation fingerprints row-by-row, and checking the executor’s
670 decision against the pre-specified rule (the “reviewer” role) — were imple-
671 mented using LLM-based coding agents operating under an orchestration
672 harness, not by hand-written deterministic scripts alone. We state this plain-
673 ly: the executor and reviewer passes described throughout Section 3.5 and in
674 the three incidents below (Appendix C.2) are automated agent passes, and
675 the incidents are, in each case, failures or near-failures of that automation.
676 We consider this disclosure a necessary complement to the pre-specification
677 protocol itself, not a caveat that weakens it: the protocol was designed on
678 the assumption that automation of this kind is fallible, and its value lies in
679 the adjudication gates of Appendix C.3, which the incidents below show
680 were load-bearing rather than decorative.

681 Appendix C.2. Three incidents

682 *Incident 1 — post-hoc criterion substitution, frozen-selectivity grid (2026-*
683 *07-03)..* The pre-specified rule for the necessity test (Section 4.1) was: at-
684 tribute robustness to selectivity only if $\Delta(\text{BM3-frozen}) \geq 10$ pp in the deep-
685 noise band, with the frozen arm collapsing toward S4D. The executor’s
686 automated decision instead wrote verdict `SELECTIVITY_CLAIM_FALSIFIED`
687 using an unpre-specified criterion — “BM3 does not significantly outper-
688 form BM3-frozen in a majority of conditions” — substituted for the nu-
689 meric threshold. On human review of the decision artifact, this substitu-
690 tion was caught before any downstream write used it: the four measured

deltas were 5.14/3.34/2.19/3.80 pp, all below the pre-specified 10 pp floor, which correctly resolves to pre-specified outcome (b), “mechanism not primary” (`SELECTIVITY_NOT_PRIMARY_MECHANISM`), a materially different and more specific claim than “falsified.” A correction card mechanically re-applied the original pre-specified rule and overwrote the erroneous decision artifact, with the human ruling and its justification timestamped in the artifact itself.

Incident 2 — post-hoc criterion substitution, component-ablation grid (2026-07-04).. During adjudication of the single-component-removal grid (Section 4.3), the executor abandoned the pre-specified percentage-point threshold and self-authored a “mean $\pm$ std interval non-overlap” heuristic, embodying it in JSON files (`secondary_prereg_20260704-0708` and `secondary_deltapp_prereg_20260704-0719`) generated *after* the grid had already run — i.e., artifacts presented as if they were the pre-specification but in fact post-dating the data they purported to govern. Separately, and independent of this substitution, the true pre-specification existed on a task-tracking card created 2026-07-03 (before the grid started at 10:34) but had not been copied into the working directory, so an independent reviewer pass, unable to see the card, correctly flagged “no pre-specification precedes this data” and proposed a full grid rerun as remedy. Human adjudication rejected the rerun: the provenance chain was intact (the true prereg predated the grid), the defect was a materialization gap, not a genuine pre-specification violation. The threshold-substitution defect was corrected by mechanically re-applying the true, now-materialized threshold (documented in `prereg_provenance.md`); the two post-hoc JSON files were not deleted but are explicitly marked void (`void_preregs`) in the corresponding decision artifact and are retained as part of the released trail (Appendix A.3). The resulting adjudication — gate removal as the only ablated component clearing the primary-driver threshold — is the one reported in Section 4.3.

Incident 3 — stale-artifact false verification, graft grid (2026-07-05).. A stage-1 task card for the graft construction (Section 4.4) was marked PASS despite no implementation work having occurred. Two compounding causes were identified on audit: (a) the card’s task section was headed `## TASK (stage-1 scope)` rather than the bare `## TASK` string the automation matched exactly, so the executor received an empty task body and did no work, logging this fact accurately; (b) the reviewer’s verification rubric described requirements generically (“scripts present”) rather than naming specific files,

727 so the reviewer verified against unrelated leftover artifacts from the prior
 728 component-ablation batch (a different grid’s unit-check and smoke-test out-
 729 puts) and passed the card without those artifacts having any bearing on the
 730 graft arm. At the time of the false PASS, the grafted arm (`s4d_plus_gate`)
 731 had no implementation anywhere in the repository — confirmed by a zero-hit
 732 search across the codebase. This was caught by an independent audit, not
 733 by the pipeline itself; the false PASS was manually voided, and the card was
 734 re-issued with three structural fixes: bare section headers only, a rubric that
 735 names exact files rather than generic requirements, and an “anti-staleness
 736 clause” requiring every verification claim to cite a file path and a modifica-
 737 tion time newer than a stated cutoff. The graft grid was then executed from
 738 scratch under the corrected card; the numbers reported in Section 4.4 and Ta-
 739 ble 5 come from that later, correctly verified run, not from the falsely-passed
 740 stage.

741 *Appendix C.3. Human adjudication gates*

742 Independent of which stages were automated, the following were, without
 743 exception, fixed or decided by the human authors, not by any agent:

744 **All numeric adjudication thresholds** (the 10 pp necessity floor in
 745 Section 4.1, the ≥ 8 pp primary-driver floor and $|\Delta| < 3$ pp not-a-driver
 746 floor in Section 4.3, the $R \geq 0.7/0.3 \leq R < 0.7/R < 0.3$ graft recovery
 747 bands in Section 4.4) were written to disk before the corresponding grid
 748 was run, as documented in Appendix A.3.

749 **All three rulings in Appendix C.2** — rejecting Incident 1’s sub-
 750 stituted criterion, rejecting Incident 2’s rerun remedy while accepting
 751 that a real (if unmaterialized) pre-specification existed, and voiding
 752 Incident 3’s false PASS — were made by the human authors on direc-
 753 t inspection of the underlying artifacts (decision JSON, task-tracking
 754 card timestamps, repository search results), not by the executor or
 755 reviewer agents that produced or reviewed those artifacts.

756 **All interpretive claims in Section 5** (the composition-bound read-
 757 ing of gating, the dual readings offered for the conv-stem cost and the
 758 graft failure) are authors’ interpretations of adjudicated numeric out-
 759 comes, not agent-generated conclusions.

760 **The decision to report, rather than suppress, all three inci-**
 761 **dents**, including the two instances where automation initially produced

762 an incorrect scientific verdict, was made by the human authors as a
763 condition of using automation in this pipeline at all.

764 No agent in this pipeline was permitted to overwrite a protected decision
765 artifact, alter a pre-specified threshold, or mark its own output as final; every
766 one of the three incidents above was caught before the erroneous artifact
767 propagated into a reported number in Section 4.

768 *Appendix C.4. Alignment with Elsevier’s generative-AI policy*

769 Elsevier’s policy on generative AI and AI-assisted technologies requires
770 that any such use in the writing process be disclosed, that AI tools not be
771 listed as authors, and that human authors retain full responsibility for the
772 content and its accuracy. The declaration accompanying this manuscript
773 (main text, “Declaration of Generative AI and AI-assisted technologies in
774 the writing process”) satisfies this requirement for the manuscript text itself.
775 We additionally extend disclosure, on our own initiative, to the experimental
776 adjudication pipeline described in this appendix, on the view that trans-
777 parency about automation matters most exactly where it could affect the
778 reported scientific conclusions, not only the prose describing them. The hu-
779 man adjudication gates of [Appendix C.3](#) are the mechanism by which we
780 take the responsibility the policy assigns to authors: every threshold, every
781 ruling on a caught defect, and every interpretive claim in this paper was
782 fixed or decided by a human author, with the automation’s role confined to
783 mechanical computation subject to audit and override.

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